

1 kHz Reference Check

Measurement Report

DATE Sunday, 12 July 2026 at 22:04
INPUT EVO4
MEASUREMENTS 1 kHz Reference, Speed & Wow Analysis

RESULT

Playback speed is slow at -0.71%; DIN-weighted wow and flutter is low at 0.045%.

MEASURED EVIDENCE

Primary Results

**Settled test-tone
frequency · Stable**

992.88 Hz
Tone frequency after settling.

**Playback speed error ·
Stable**

-0.71 %
Playback speed is slow.

Calculated RPM · Stable

33.10 RPM
Measured under stylus load.

**Settled RPM range ·
Stable**

32.63 – 33.55 (span 0.93) RPM
Settled minimum, maximum, and span.

**DIN-weighted wow &
flutter · Stable**

0.045 %
Low.

**Raw speed variation ·
Stable**

RMS ± 0.358 · $2\sigma \pm 0.716$ %
Unweighted diagnostic; do not compare directly with DIN
thresholds.

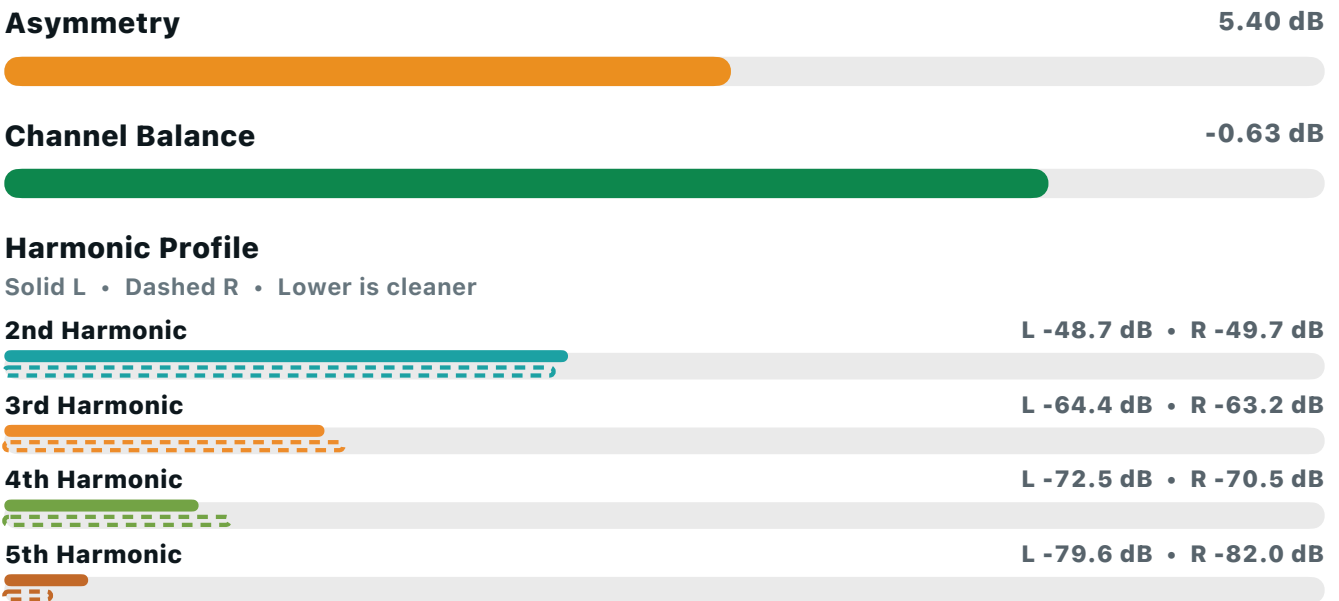
**Raw wow / flutter RMS ·
Stable**

0.186 / 0.287 %
Unweighted diagnostic components.

Supporting Diagnostics

1 kHz channel balance · Diagnostic only	-0.63 (moderate imbalance) dB A channel-level difference was observed. Confirm it before alignment changes.
Dedicated 1 kHz crosstalk · Supporting	Average 30.4 • L→R -27.7 • R→L -33.1 • Δ 5.40 dB Average separation is the main 1 kHz value; directional difference requires repetition.
1 kHz distortion · Diagnostic only	L 0.37 • R 0.34 • L-R +0.03 % Left/right THD is variable between runs; repeat before setup changes.
Harmonic character · Diagnostic only	H2 L -48.7 dB / R -49.7 dB / Δ +0.9 dB H3 L -64.4 dB / R -63.2 dB / Δ -1.2 dB H4 L -72.5 dB / R -70.5 dB / Δ -2.0 dB H5 L -79.6 dB / R -82.0 dB / Δ +2.4 dB The harmonic differences are isolated or change direction, so no consistent channel signature is established. Harmonic differences should influence setup only when the same multi-harmonic direction repeats across captures.
Combined 1 kHz THD · Supporting	0.31 % Computed from the combined mono 1 kHz captures, not from the arithmetic mean of left/right THD.
Relative residual noise · Diagnostic only	14.8 dB Referenced to the extracted fundamental; not equivalent to an A-weighted bench noise measurement.

DIAGNOSTIC BALANCE



INTERPRETATION

Playback speed and rotational stability were measured consistently during this capture.

Channel-specific distortion, harmonic, balance, and directional 1 kHz measurements are supporting diagnostics and require repetition before mechanical adjustment.

NOTES

Completed measurements: 1 kHz Reference.

RECOMMENDATION

Confirm the speed result with another valid capture, then correct the deck speed if the same error repeats. Single-session observation — repeat variable diagnostics before adjusting anti-skate, zenith, azimuth, VTA, or VTF.